Heat Flow Across a Material Interface

Conservative FDM and piecewise-linear FEM

MTH321 Project 2

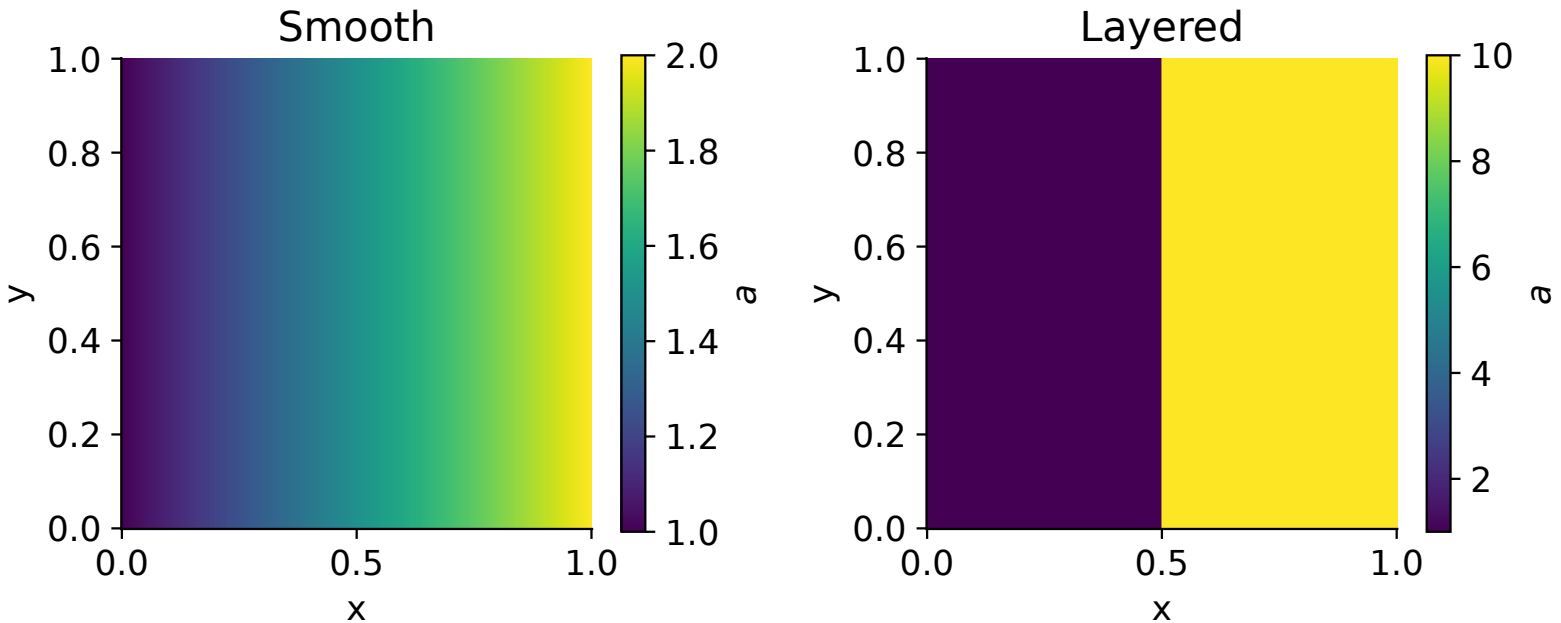
XJTLU

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What must a numerical method preserve when conductivity jumps?

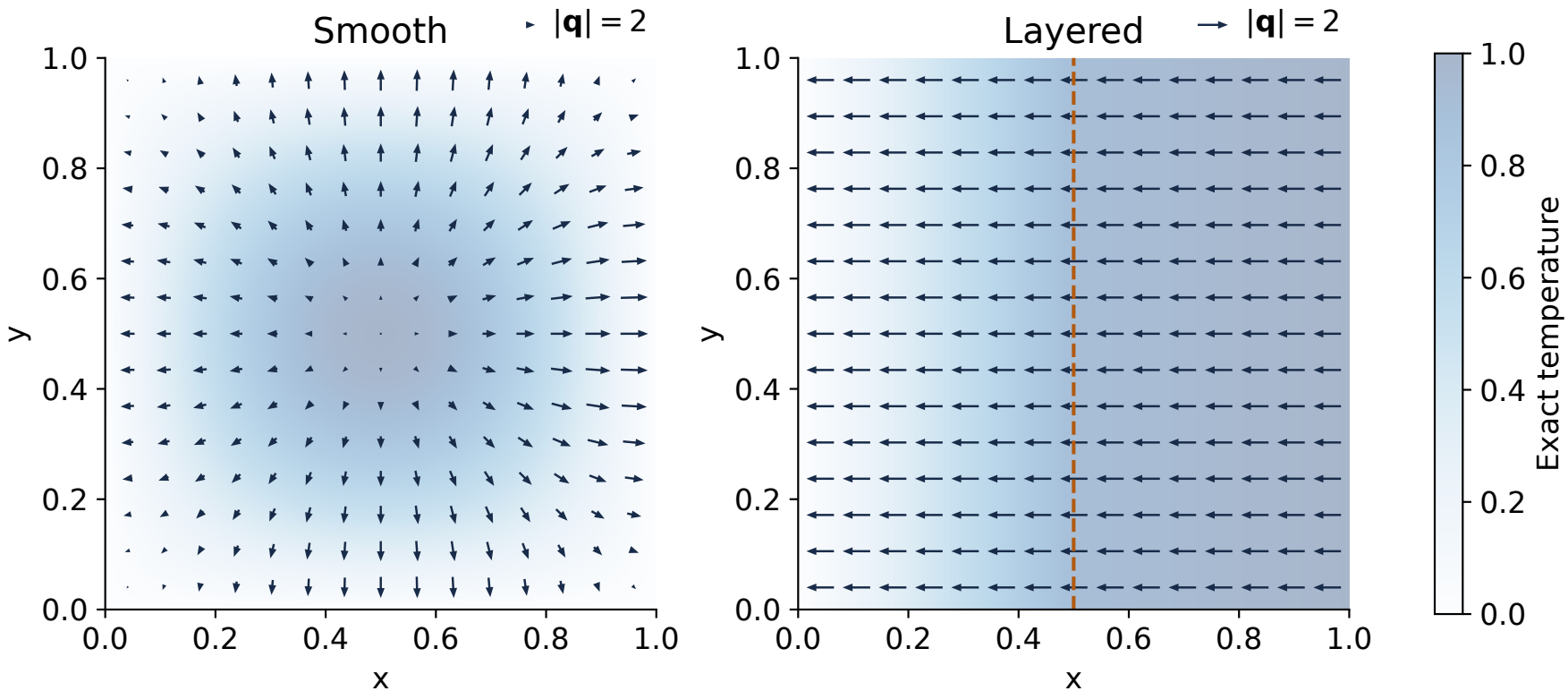
Two conductivity models

$$-\nabla \cdot (a \nabla u) = f, \quad \mathbf{q} = -a \nabla u, \quad u = g \text{ on } \partial\Omega.$$



On $\Omega = (0,1)^2$: smooth $a = 1 + x$ tests accuracy; layered $a_L = 1$, $a_R = 10$ tests interface physics. Colors show conductivity; the two color scales differ.

Heat flux in the two models



Background: exact temperature. Arrows: exact $\mathbf{q} = -a\nabla u$. Each panel has its own arrow key; compare magnitudes through those keys.

The layered flux points left and is constant across the interface $x = 1/2$.

What the interface must preserve

For perfect contact with no interface source,

$$[u] = 0, \qquad [\mathbf{q} \cdot \mathbf{n}] = 0.$$

The layered benchmark has $f = 0$ and exact solution

$$u^*(x) = \begin{cases} 20x/11, & x \le 1/2, \\ (2x+9)/11, & x \ge 1/2. \end{cases}$$

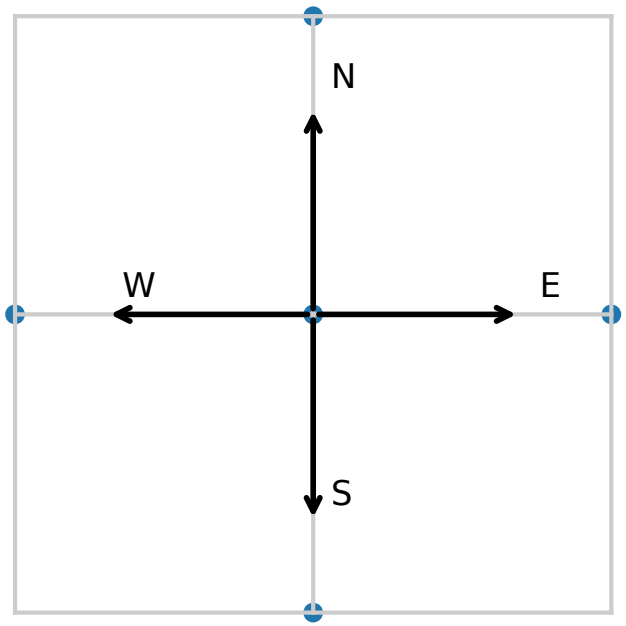
$$u_x^- = 20/11, \quad u_x^+ = 2/11, \quad q_x^* = -20/11.$$

Temperature is continuous; its slope changes so that flux stays continuous.

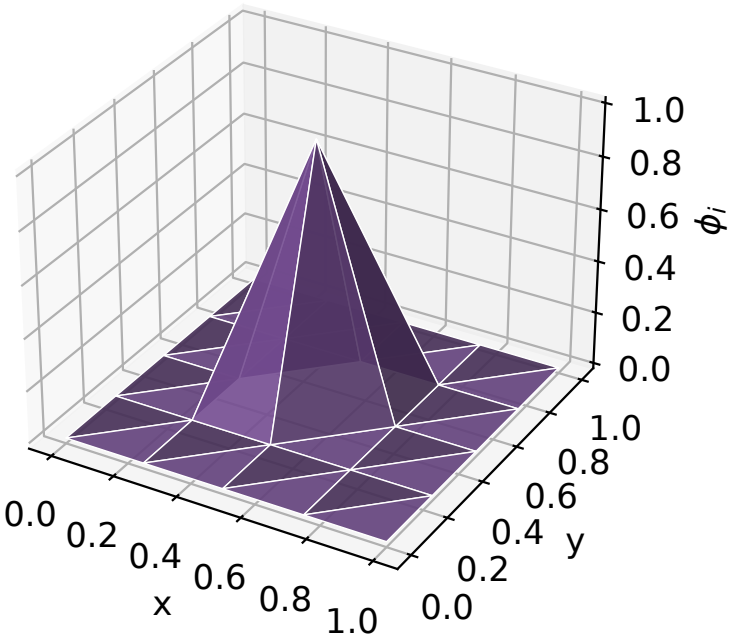
We impose $g = u^*$ on **all four boundary sides**.

Two numerical representations

FDM: shared face fluxes



FEM: P1 nodal basis



FDM: balance fluxes shared by neighboring nodes.
FEM: combine nodal hat functions, $u_h = \sum_i U_i \phi_i$.

Left arrows indicate stencil directions. The P1 basis function on the right is one at its own node, zero at the other nodes, and linear on each triangle.

FDM: coefficients and boundary assembly

$$q_{i+1/2,j} = -a_{i+1/2,j}\frac{U_{i+1,j} - U_{i,j}}{h}.$$

Taking the discrete divergence gives

$$(A_h U)_P = \frac{1}{h^2} \sum_{s=E,W,N,S} a_s(U_P - U_s).$$

- Smooth coefficient: $a_{i+1/2,j} = (a_{i,j} + a_{i+1,j})/2$.
- Share the same a_s in the two adjacent rows.
- For a boundary neighbor, $U_s = g_s$: add $a_s g_s/h^2$ to the right-hand side.

Positive shared weights and boundary elimination yield a symmetric positive-definite system.

FEM: coefficients and boundary assembly

For test functions $v_h = 0$ on the boundary,

$$\int_\Omega a \nabla u_h \cdot \nabla v_h = \int_\Omega f v_h.$$

Write $u_h = \sum_j U_j \phi_j$ and assemble element contributions:

$$K_{ij} = \sum_T \int_T a \nabla \phi_j \cdot \nabla \phi_i, \qquad F_i = \sum_T \int_T f \phi_i.$$

Separate interior and boundary degrees of freedom:

$$K_{II} U_I = F_I - K_{IB} g_B.$$

- Smooth a: quadrature integrates coefficient and load terms.
- Layered a: fitted elements lie within one material each.

Positive weak energy gives SPD after Dirichlet elimination.

How we measure error

Average error over the domain:

$$E_{L^2} = \left(\int_\Omega |u^* - u_h|^2 \, dx \right)^{1/2}.$$

Largest error at the sampled nodes:

$$E_{\infty,\text{nodes}} = \max_i |u^*(x_i) - U_i|.$$

FDM: integrate a reconstructed function; FEM: integrate the P1 function. The nodal maximum is a discrete L^∞ measure, not the continuous supremum.

For successive mesh sizes $h, \widetilde{h}$,

$$p = \frac{\log(E_h / E_{\widetilde{h}})}{\log(h/\widetilde{h})}.$$

Why smooth FDM is second order

At a face midpoint m,

$$\frac{u(x+h)-u(x)}{h} = u'(m) + \frac{h^2}{24}u'''(m) + O(h^4).$$

The coefficient average also approximates $a(m)$ to order h^2. Taking the flux difference gives the consistency defect

$$\tau_h = b_h - A_h R_h u^* = O(h^2).$$

The nodal error satisfies

$$A_h(U_h - R_h u^*) = \tau_h.$$

Consistency + stability gives second-order solution error for this smooth, uniformly elliptic problem.

Why P1 FEM is second order in L^2

Let $B(w,v) = \int_\Omega a\nabla w \cdot \nabla v$ and $e = u - u_h$. Galerkin orthogonality gives $B(e, v_h) = 0$.

$$\|e\|_a \le \|u - I_h u\|_a \le Ch|u|_{H^2}.$$

For the dual problem $B(v,z) = (v,e)$,

$$\|e\|_{L^2}^2 = B(e, z - I_h z) \le Ch\|e\|_a \|z\|_{H^2}.$$

With dual regularity $\|z\|_{H^2} \le C\|e\|_{L^2}$,

$$\boxed{\|e\|_{L^2} \le Ch^2}, \qquad \|e\|_a = O(h).$$

This does not by itself prove an unconditional $O(h^2)$ maximum-norm bound.

Separating numerical error sources

$$\text{total error} = \text{discretization} + \text{quadrature} + \text{algebraic error}.$$

The smooth benchmark uses

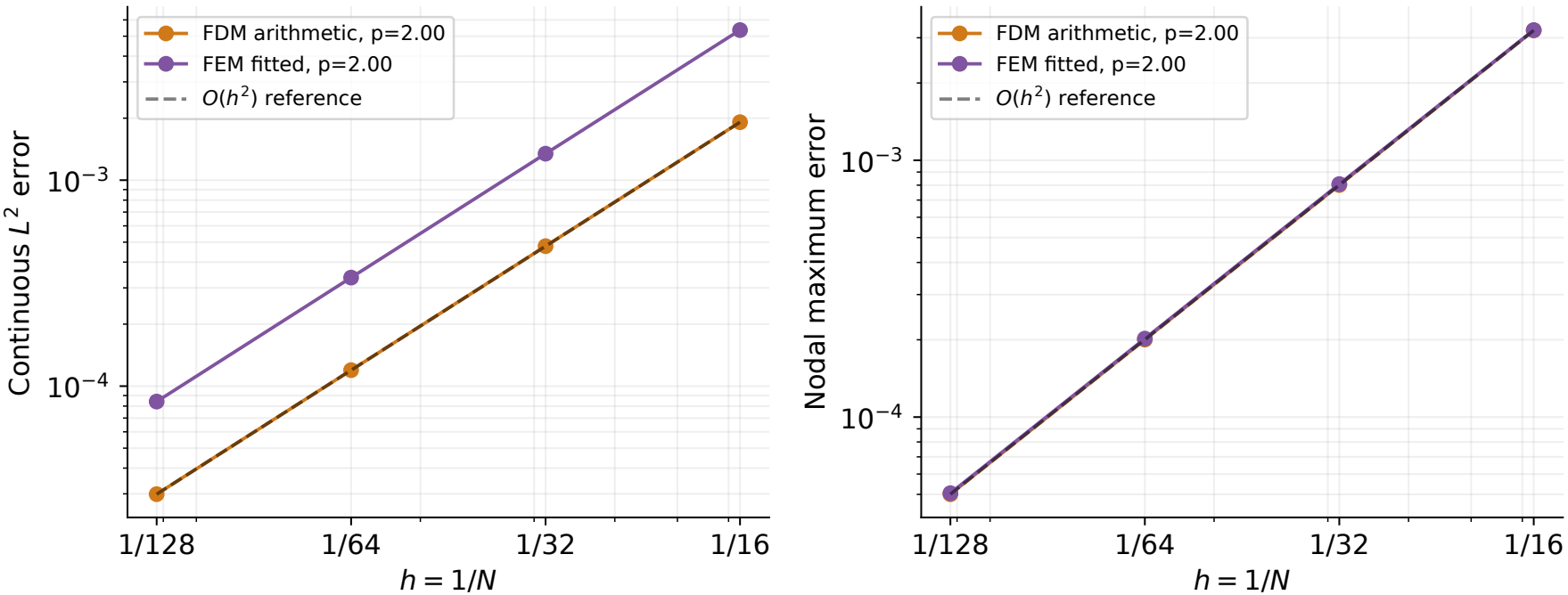
$$u^* = \sin(\pi x)\sin(\pi y), \quad a = 1 + x, \quad g = 0,$$

$$f = 2\pi^2(1 + x)\sin(\pi x)\sin(\pi y) - \pi\cos(\pi x)\sin(\pi y).$$

- Independently check f by differentiating the analytic flux numerically.
- Estimate algebraic error through a residual-correction solve.
- Compare FEM quadrature orders 6 and 8.

Both extra effects are below 10^{-3} of the smooth measured L^2 error.

Smooth benchmark: convergence in both measures



Four meshes: $N = 16, 32, 64, 128$, with $h = 1/N$. Left: continuous L^2; right: nodal maximum. Dashed black lines indicate slope two; fitted orders use the finest three meshes.

Both methods approach second order here; error constants differ.

Why the interface needs a different argument

At $x = 1/2$, a and u_x jump. A Taylor expansion through that point is invalid.

The physically useful invariant is instead

$$q_x = -a u_x = \text{constant}.$$

- **FDM**: what coefficient gives the correct resistance across an edge?
- **FEM**: can the approximation space represent the slope change?

The two methods answer these questions differently.

FDM: derive the interface resistance

Integrate $u_x = -q_x/a$ along an edge crossing the interface:

$$u_R - u_L = -q_x \left(\frac{d_L}{a_L} + \frac{d_R}{a_R} \right).$$

Match this to $q_x = -a_{\text{edge}}(u_R - u_L)/h$, where $h = d_L + d_R$:

$$\boxed{a_{\text{edge}} = \frac{h}{d_L/a_L + d_R/a_R}}.$$

For a bisected edge,

$$a_H = \frac{2a_L a_R}{a_L + a_R} = \frac{20}{11}.$$

The two material resistances add in series.

Why arithmetic averaging fails here

For the same edge,

$$a_A = \frac{1+10}{2} = 5.5, \qquad a_H = \frac{20}{11} \approx 1.818.$$

Inserting the *exact nodal temperature drop* into each flux formula gives

$$\frac{q_A[u^*]}{q^*} = \frac{a_A}{a_H} = \frac{121}{40}.$$

- Arithmetic averaging assigns too little thermal resistance.
- The computed temperature adjusts when the full system is solved.
- 121/40 is a consistency check, not the ratio of re-solved fluxes.

Positive coefficients can still give symmetry and boundedness while encoding the wrong interface physics.

FEM: why fit the interface?

On each triangle, P1 FEM has a constant gradient.

- A triangle crossing the interface cannot represent both slopes.
- A fitted mesh places the kink on element edges.
- Integrate a within each material; impose exact boundary data.

For this benchmark, u^* is continuous and affine within each material:

$$u^* \in V_h, \qquad B(u^* - u_h, v_h) = 0.$$

Choosing $v_h = u^* - u_h$ gives

$$\|u^* - u_h\|_a^2 = 0.$$

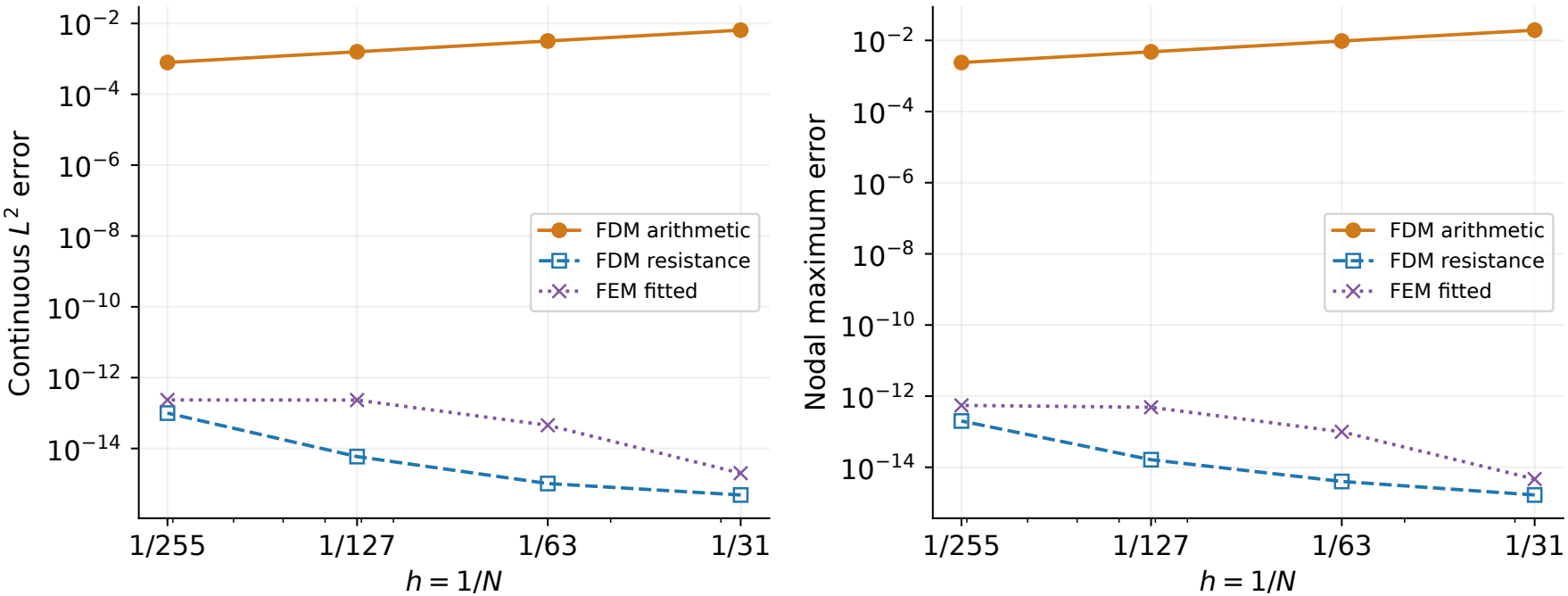
Hence fitted FEM reproduces this solution in exact arithmetic.

Three methods in the layered comparison

Method	Interface treatment	Plot identity
FDM arithmetic	Arithmetic face coefficient	Orange circles; solid line
FDM resistance	Subedge resistance; harmonic for equal lengths	Blue hollow squares; dashed line
FEM fitted	Interface on triangle edges; material-wise integration	Purple crosses; dotted line

FDM uses odd $N = 31, 63, 127, 255$; FEM inserts the interface column. Macro spacing is comparable, but unknown counts differ.

Layered refinement: L^2 and nodal maximum error



As for the smooth model: continuous L^2 error on the left, nodal maximum error on the right, with $h = 1/N$ on four meshes.

Arithmetic errors decrease; the other two methods already have tiny errors that do not follow the smooth model's convergence pattern. Why?

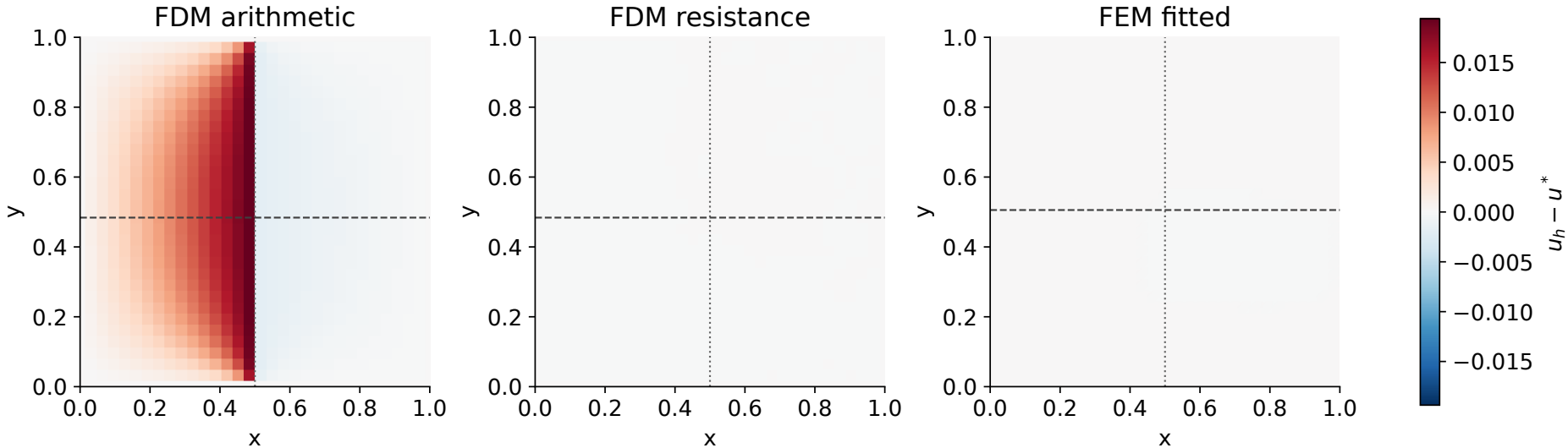
Why the usual convergence test is inconclusive here

This benchmark is **exactly representable**, rather than simply very smooth:

- u^* is affine within each layer, with a kink at the interface.
- Resistance FDM gives the exact nodal solution; fitted P1 FEM contains u^*.
- Their discretization error is zero in exact arithmetic. Floating-point computation and the linear solve leave small measured errors.

We cannot infer a general convergence order from those tiny errors. The benchmark instead tests whether the interface is represented correctly.

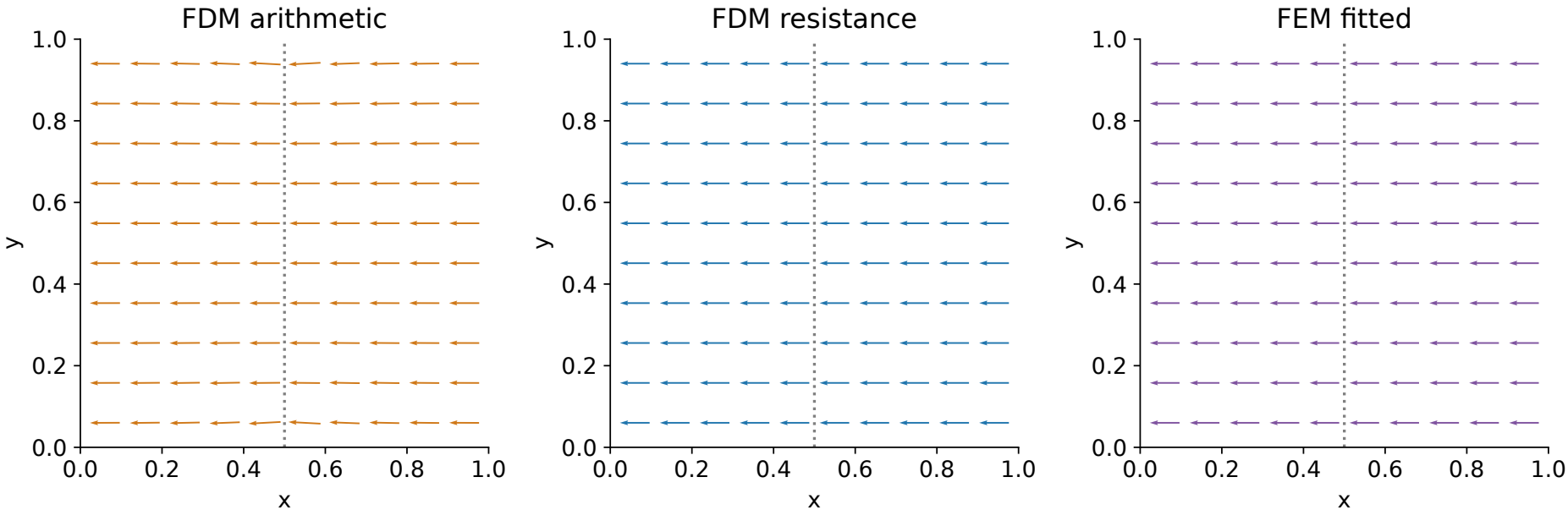
Where does the layered temperature error occur?



Signed nodal error $u_h - u^*$ at $N = 31$, with one shared color scale. Vertical dotted lines mark the interface; horizontal dashed lines mark the slices used below.

Arithmetic error concentrates around the interface and varies with y. Resistance FDM and fitted FEM are essentially zero on this scale.

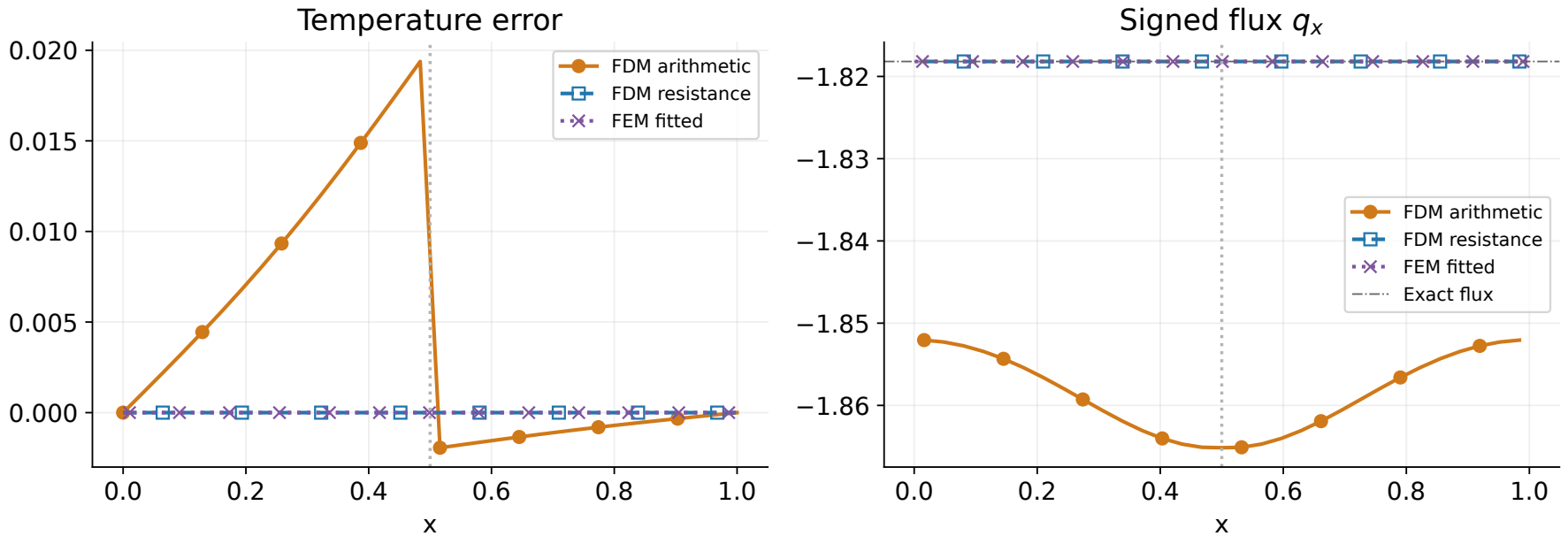
Inspect the flux field before taking a slice



Arrows show numerical $\mathbf{q}_h$, enlarged with the same scale for all methods; lengths are not normalized. Dotted lines mark $x = 1/2$.

The exact field is $\mathbf{q}^* = (-20/11, 0)$: horizontal and independent of y. Arithmetic FDM introduces small changes in both magnitude and direction.

Horizontal slices reveal the signed flux error



Along $y = y_0$, plot $u_h - u^*$ and the interface-normal component $q_{x,h}(x, y_0)$. FDM: $y_0 = 0.484$; FEM: $y_0 = 0.506$; $N = 31$. The gray reference is $q_x^* = -20/11$; the dotted vertical line is the interface.

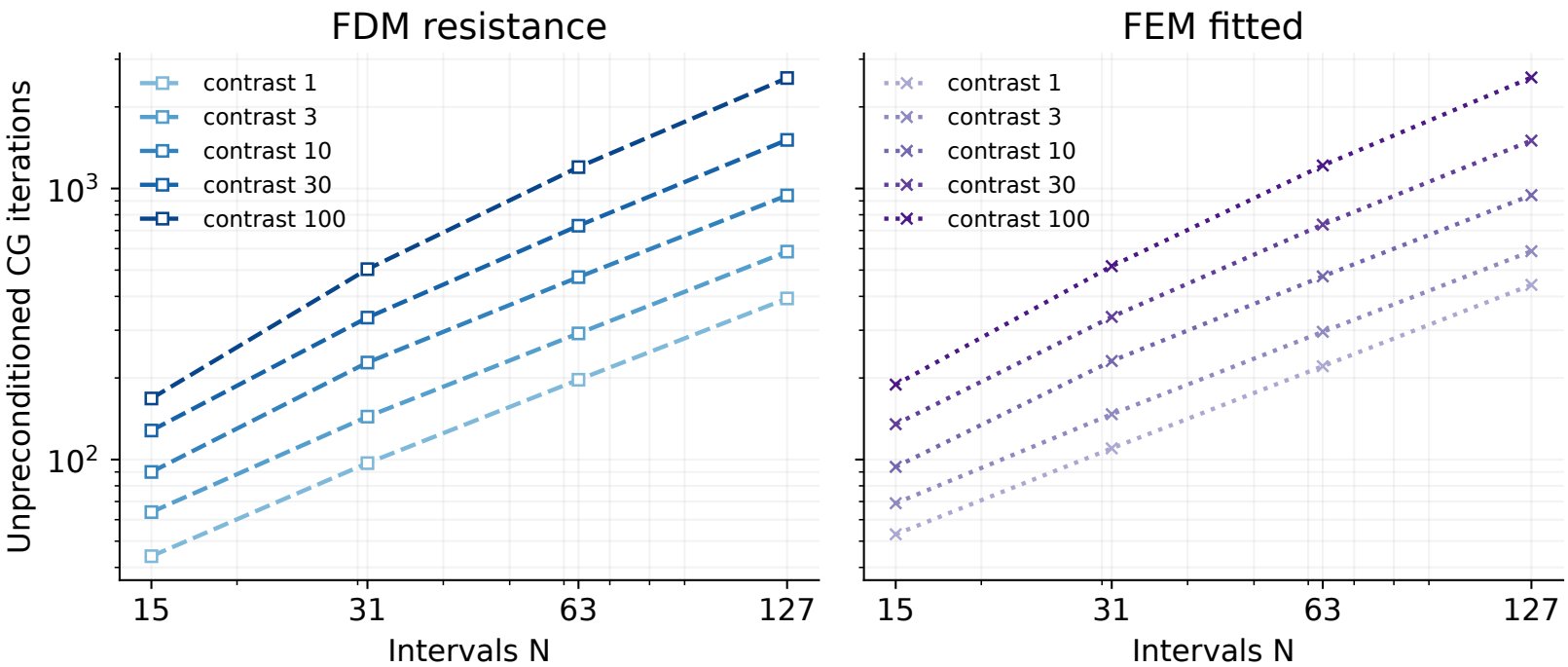
This slice checks normal flux; it does not replace the full 2D field. Blue squares and purple crosses recover the reference. Arithmetic FDM does not.

What validates the interface result?

- Compare signed flux with the exact $-20/11$.
- Reconstruct fluxes independently on the two material sides; sharing one face flux is not an independent continuity check.
- All layered solutions remain between boundary extrema 0 and 1.
- Positive diagonal and nonpositive off-diagonal entries support a discrete maximum principle on these meshes.

Bounded temperature does *not* prove accuracy: arithmetic FDM passes that check too.

Refinement and contrast increase solver work



Unpreconditioned CG: vary mesh resolution N and contrast a_R/a_L. Blue: resistance FDM; purple: fitted FEM. Darker shades mean higher contrast; both axes are logarithmic.

Plotted values are iteration counts, with zero initial guess and $\|b - AU\|_2/\|b\|_2 \leq 10^{-10}$ in every case. Counts show increasing difficulty in the tested cases.

Why?

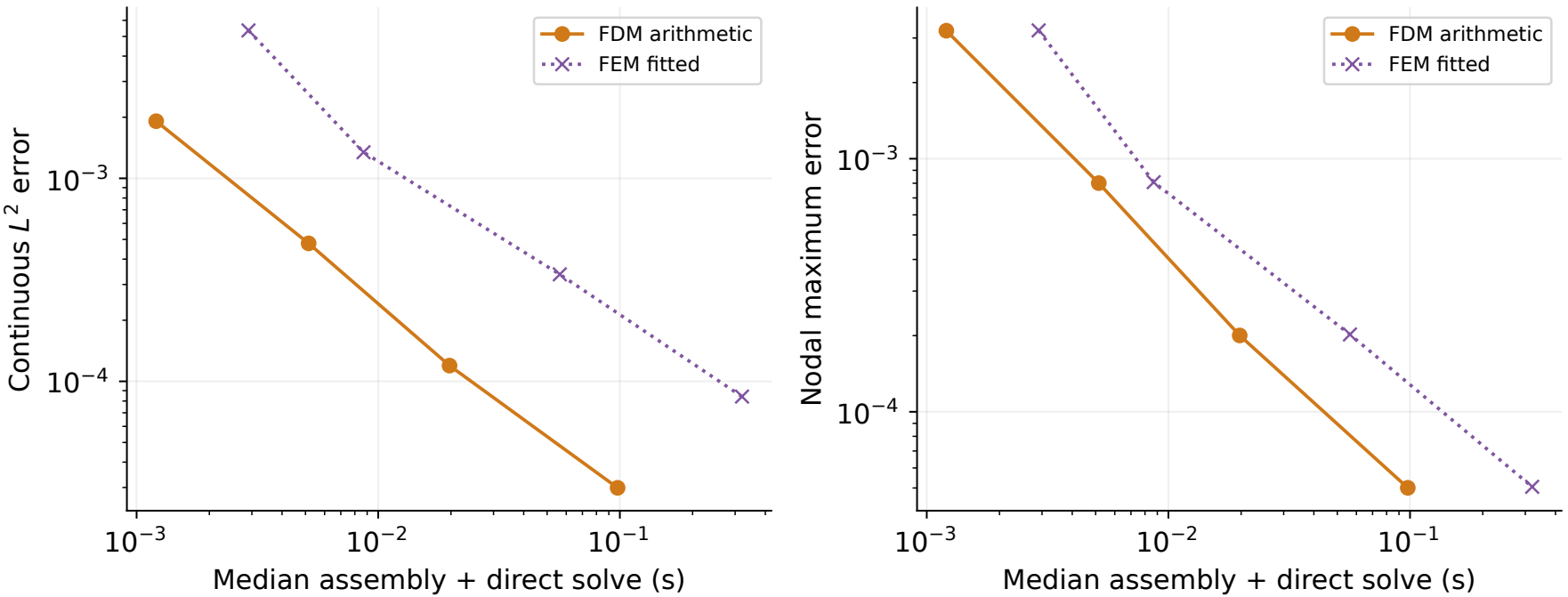
Cost versus accuracy: what is measured?

$$T = T_{\text{assembly}} + T_{\text{direct solve}}.$$

- One warm-up, then three timed repetitions per case; plot the median.
- Exclude error diagnostics, plotting and file I/O from the timing.
- Use verified errors and the same mesh cases as the refinement studies.
- Compare these implementations; FEM and FDM assembly costs differ.

In 2D, unknown count and sparse assembly/storage scale as $O(h^{-2})$. Sparse direct factorization can scale as $O(h^{-3})$ with suitable ordering; actual timings depend on implementation and hardware.

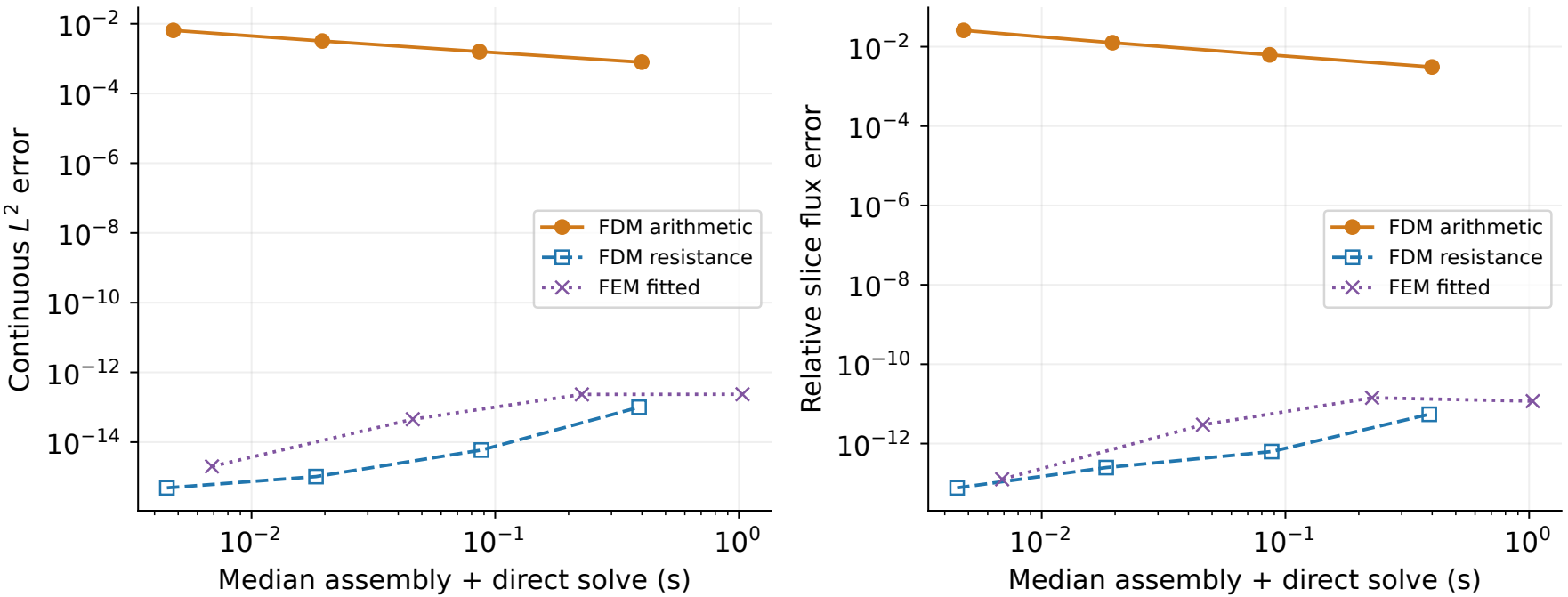
Smooth model: cost versus accuracy



Horizontal axis: median assembly plus direct-solve time in seconds. Vertical axes: continuous L^2 and nodal maximum errors. Each marker is one mesh; connecting lines follow refinement.

FDM attains lower errors at lower measured cost in these runs. This describes these implementations and this benchmark.

Layered model: cost versus accuracy



Left: continuous L^2 error. Right: maximum relative signed-flux error along the sampled slice, $\max |q_{x,h} - q_x^*| / |q_x^*|$. Time includes assembly plus direct solve, using the same timing protocol.

Refinement reduces the arithmetic error. Exact representations already reach numerical precision; added work does not give a comparable reduction.

Conclusions and task coverage

Task / comparison	Work completed and main finding
1. Assembly and SPD	**Complete; partially presented.** Conservative assembly, Dirichlet elimination and matrix checks.
2. Smooth verification	**Completed and reported.** Both methods approach second-order L^2 accuracy; both error norms compared.
3. Interface comparison	**Completed and reported.** Resistance FDM and fitted FEM reproduce the exact solution.
4. Boundary extrema	Bounds checked: $0 \leq U \leq 1$. Boundedness alone does not establish accuracy.
5. Solver work	**Completed and reported.** Refinement and contrast increase CG counts at a matched residual tolerance.
(Optional) FDM–FEM	**Completed.** Both models compared at matched mesh scale using common error measures.
(Optional) cost–accuracy	**Completed.** Assembly plus direct-solve cost compared with solution and flux errors.

Not developed here / open question: Why do the matrix signs keep the layered solution within its boundary extrema?